ROC Curve & Precision-Recall Curve

Receiver Operating Characteristic (**ROC**) curves for a binary classifier:

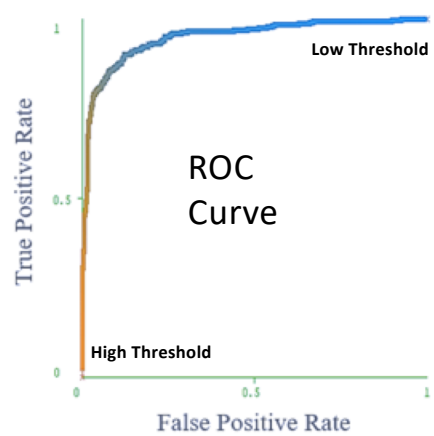
True Positive Rate (TPR) vs **False Positive Rate (FPR)**
at various threshold settings.

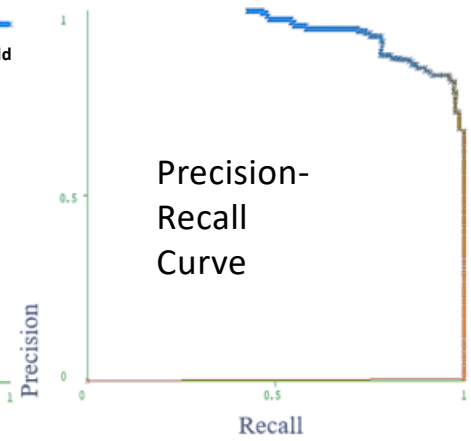
ROC curve was first developed by radar engineers during World War II for detecting enemy objects.

TPR = REC = **Recall or Sensitivity** = TP / (TP + FN)
(planes classified as planes / all planes' events)

FPR = fall-out, probability of false alarm = FP / (FP + TN)
(noise classified as planes / all noise events)

PRE = **Precision** = TP/(TP+FP)
(planes classified as planes / all events classified as planes)





AUC = Area Under the Curve

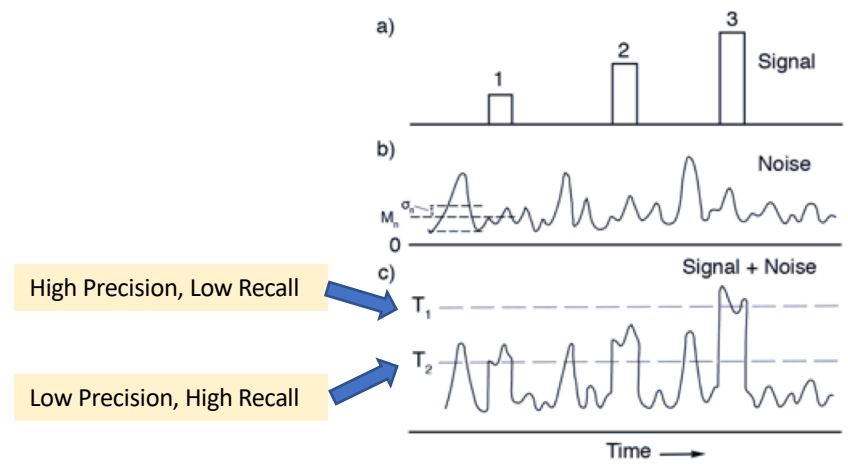
Simply the area under the ROC curve (from 0 to 1)

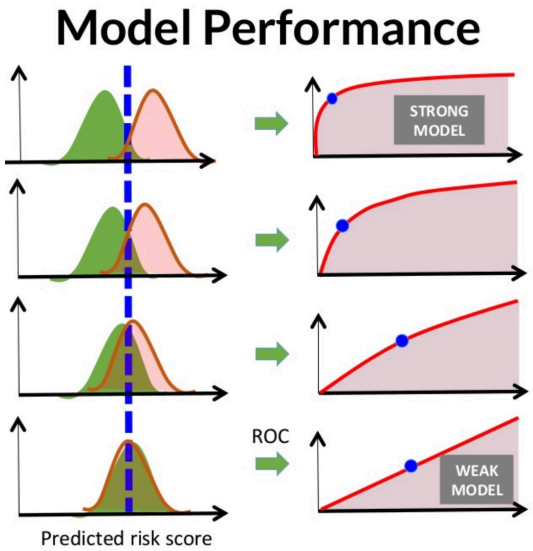
F1 Score

F1 score is a single number evaluation metric:

F1 score = 2 * PRE * REC / (PRE + REC)

So the **F1 score** is the harmonic average of the **precision** and **recall**.





From:
https://www.slideshare.net/KristenHunter/civitas-learning-understanding-roc-curves

For measuring the performance of retrieval systems, several measures are used that depends on the following four major parameters:

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True Positive  (TP): True  is identified True  (correct identification)
True Negative  (TN): False is identified False (correct identification)
False Positive (FP): True  is identified False (wrong identification)
False Negative (FN): False is identified True  (wrong identification)

P = TP + FN (number of correct identification cases)
N = FP + TN (number of wrong identification case) 25

Sensitivity or True Positive Rate (TPR)       : TPR = (TP / P) = TP / (TP + FN)
Specificity (SPC) or True Negative Rate (TNR) : SPC = (TN / N) = TN / (FP + TN)
Precision or Positive Predictive Value (PPV)  : PPV = TP / (TP + FP)
Negative Predictive Value (NPV)               : NPV = TN / (TN + FN)
Fall-Out or False Positive Rate (FPR)         : FPR = (FP / N) = FP / (FP + TN) = 1 – SPC
False Discovery Rate (FDR)                    : FDR = FP / (FP + TP) = 1 – PPV
Miss Rate or False Negative Rate (FNR)        : FNR = FN / (FN + TP)
Accuracy (ACC)                                : ACC = (TP+TN) / (P+N)
F1 Score                                      : F1  = 2*TP/(2*TP+FP+FN)

Mathews Correlation Coefficient (MCC):
    MCC=(TP*TN-FP*FN)/sqrt((TP+FP)*(TP+FN)*(TN+FP) *(TN+FN))

Informedness = Sensitivity + Specificity -1
Markedness   = Precision + NPV -1
```

From:
https://www.researchgate.net/post/How_do_I_calculate_the_false_alarm_rate_for_face_detection

Confusion Matrix

For a simple binary classifier we are predicting one of two possibilities (Yes/No, Positive/Negative, Plane/No-Plane, Normal/Abnormal, etc.).

So we construct a 2x2 matrix to visualize True & False Positives and Negatives.

		Predicted class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

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from sklearn import metrics

expected = target
predicted = model.predict(data)

confusion = metrics.confusion_matrix(expected, predicted)

[[194156   4864]
 [    23    321]]
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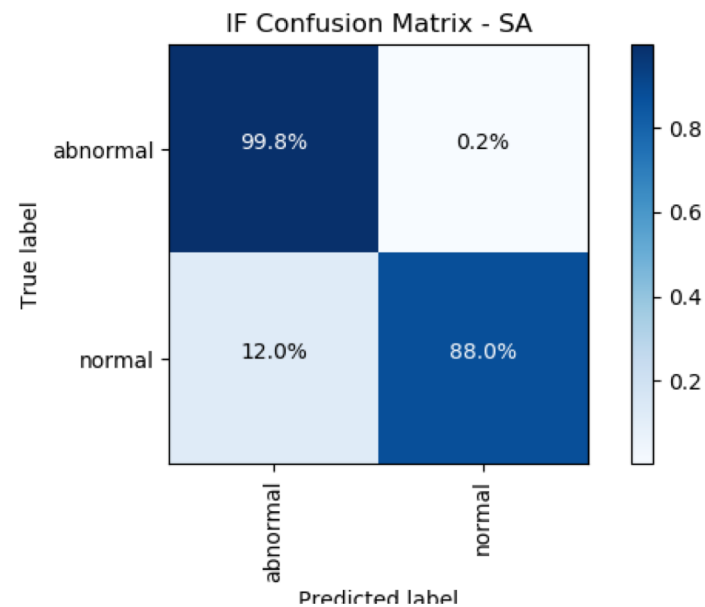
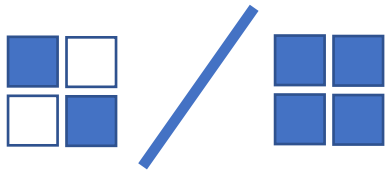


Accuracy = sum(Diagonal) / sum(ALL)

Precision = TP / (TP+FP)

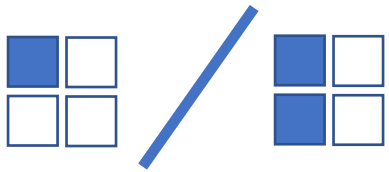
Recall = TP / (TP+FN) = TP / All_Actual_Positives

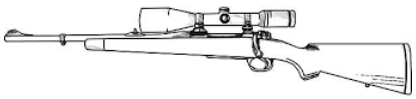
Accuracy = sum(Diagonal) / sum(ALL)



		Predicted class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

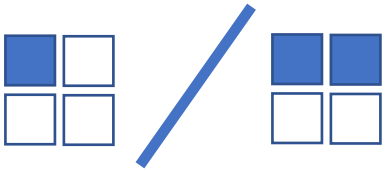
Precision = TP / (TP+FP) = TP / Predicted_Positives





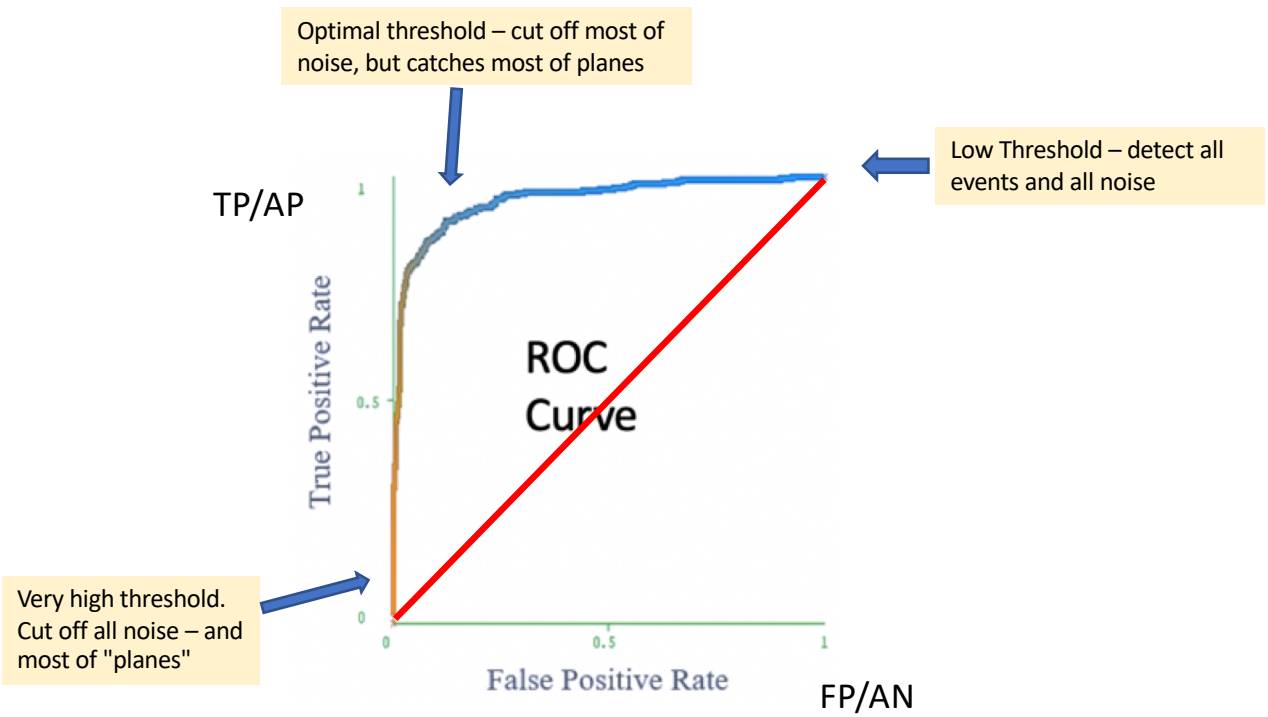
Sniper
- high precision (zero FP)
- low recall (can't get everything)

Recall = TP / (TP+FN) = TP / Actual_Positives

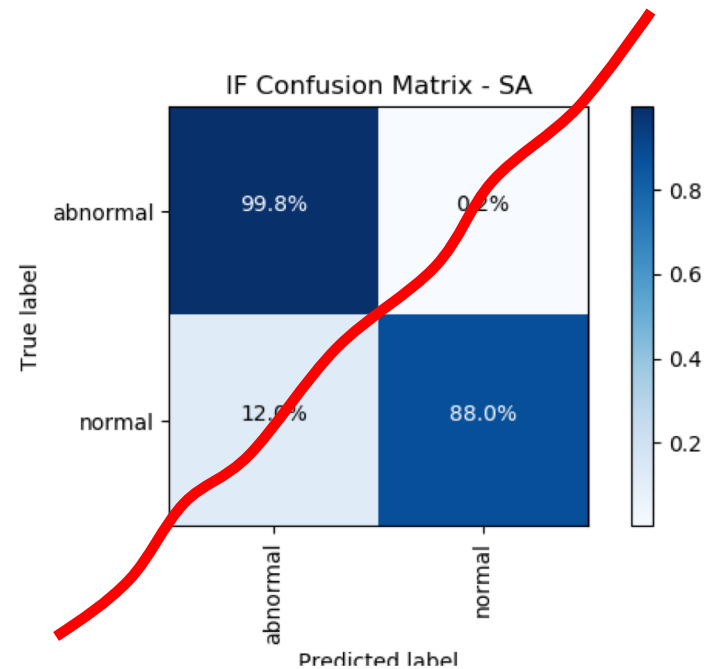


Nuclear Bomb
- low precision (many FP)
- high recall (get everything)

Why good ROC curve is above diagonal



		Predicted class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)



ROC curve axes:

vertical – True Positives / All Actual Positives

horizontal – False Positives / All Actual Negatives